DAA LAB ASSIGNMENT WEEK-3

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1) BFS

CODE:

```
#include <stdio.h>

#define MAX 20

int graph[MAX][MAX];
int visited[MAX];
int queue[MAX];
int front=-1,rear=-1;
int n;

void enqueue(int x){
   if(rear==MAX-1)
      return;
   if(front==-1)
      front=0;
   queue[++rear]=x;
}

int dequeue(){
   if(front==-1 || front>rear)
```

```
        return -1;
    return queue[front++];
}

void bfs(int start){
    int i;
    visited[start]=1;
    enqueue(start);
    printf("\nBFS Traversal:\n");
    while(front!=-1 && front<=rear){
        int current=dequeue();
        printf("%d ",current+1);
        for(i=0;i<n;i++){
            if(graph[current][i]==1 && !visited[i]){
                visited[i]=1;
                enqueue(i);
            }
        }
    }
    printf("\n");
}

void displayMatrix(){
    int i,j;
    printf("\nAdjacency Matrix:\n    ");
    for(i=0;i<n;i++)
        printf("%d ",i+1);
    printf("\n   ");
```

```
    for(i=0;i<n;i++)
        printf("--");
    printf("\n");
    for(i=0;i<n;i++){
        printf("%d | ",i+1);
        for(j=0;j<n;j++)
            printf("%d ",graph[i][j]);
        printf("\n");
    }
}

int main(){
    int i,j,start,edge;
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            graph[i][j]=0;
    printf("\nEnter edges (1 for yes, 0 for no):\n");
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            printf("Is there an edge between %d and %d? ",i+1,j+1);
            scanf("%d",&edge);
            if(edge==1){
                graph[i][j]=1;
                graph[j][i]=1;
            }
        }
```

```
    }
    displayMatrix();
    for(i=0;i<n;i++)
        visited[i]=0;
    printf("\nEnter starting vertex for BFS (1-%d): ",n);
    scanf("%d",&start);
    bfs(start-1);
    return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./bfs
Enter number of vertices: 5

Enter edges (1 for yes, 0 for no):
Is there an edge between 1 and 2? 1
Is there an edge between 1 and 3? 0
Is there an edge between 1 and 4? 0
Is there an edge between 1 and 5? 1
Is there an edge between 2 and 3? 1
Is there an edge between 2 and 4? 0
Is there an edge between 2 and 5? 0
Is there an edge between 3 and 4? 0
Is there an edge between 3 and 5? 1
Is there an edge between 4 and 5? 1

Adjacency Matrix:
    1 2 3 4 5
   ---------
1 | 0 1 0 0 1
2 | 1 0 1 0 0
3 | 0 1 0 0 1
4 | 0 0 0 0 1
5 | 1 0 1 1 0

Enter starting vertex for BFS (1-5): 1

BFS Traversal:
1 2 5 3 4
```

2) DFS

CODE:

```
#include <stdio.h>

#define MAX 20

int graph[MAX][MAX];
int visited[MAX];
int n;

void dfs(int node){
   int i;
   visited[node]=1;
   printf("%d ",node+1);
   for(i=0;i<n;i++){
      if(graph[node][i]==1 && !visited[i]){
         dfs(i);
      }
   }
}

void displayMatrix(){
   int i,j;
   printf("\nAdjacency Matrix:\n    ");
   for(i=0;i<n;i++)
      printf("%d ",i+1);
   printf("\n   ");
   for(i=0;i<n;i++)
```

```
        printf("--");
    printf("\n");
    for(i=0;i<n;i++){
        printf("%d | ",i+1);
        for(j=0;j<n;j++)
            printf("%d ",graph[i][j]);
        printf("\n");
    }
}

int main(){
    int i,j,start,edge;
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            graph[i][j]=0;
    printf("\nEnter edges (1 for yes, 0 for no):\n");
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            printf("Is there an edge between %d and %d? ",i+1,j+1);
            scanf("%d",&edge);
            if(edge==1){
                graph[i][j]=1;
                graph[j][i]=1;
            }
        }
    }
```

```
    displayMatrix();
    for(i=0;i<n;i++)
        visited[i]=0;
    printf("\nEnter starting vertex for DFS (1-%d): ",n);
    scanf("%d",&start);
    printf("\nDFS Traversal:\n");
    dfs(start-1);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ ./dfs
Enter number of vertices: 5

Enter edges (1 for yes, 0 for no):
Is there an edge between 1 and 2? 1
Is there an edge between 1 and 3? 0
Is there an edge between 1 and 4? 0
Is there an edge between 1 and 5? 1
Is there an edge between 2 and 3? 1
Is there an edge between 2 and 4? 0
Is there an edge between 2 and 5? 0
Is there an edge between 3 and 4? 0
Is there an edge between 3 and 5? 1
Is there an edge between 4 and 5? 1

Adjacency Matrix:
    1 2 3 4 5
   ----------
1 | 0 1 0 0 1
2 | 1 0 1 0 0
3 | 0 1 0 0 1
4 | 0 0 0 0 1
5 | 1 0 1 1 0

Enter starting vertex for DFS (1-5): 1

DFS Traversal:
1 2 3 5 4
```